SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 10%

QUESTIONS: 15 **Total Marks: 25**

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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ITA0606 - Machine Learning

Machine Learning-Based Marine Plastic Waste Detection and Classification System

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Abstract

- Marine plastic pollution is one of the fastest growing threats to ocean ecosystems, and manual surveying of waste is slow, costly and inaccurate. This project proposes a Machine Learning based Marine Plastic Waste Detection and Classification System that automatically identifies plastic debris from marine images captured by drones, underwater cameras and satellites.
- **Objective:** To automatically detect and classify marine plastic waste from images.
- **Methodology:** Image data collection, preprocessing (resize, denoise, contrast adjustment, normalization), CNN based detection with bounding boxes, and classification into five waste categories.
- **Key Results:** The trained model achieves high detection accuracy with real-time inference and presents results through a monitoring dashboard that supports marine cleanup planning.

Problem Statement

- Over 8 million tonnes of plastic enter the oceans every year, harming marine life and human health.
- Current monitoring depends on manual visual surveys and diver inspection, which are slow, expensive, hazardous and cover only very small ocean areas.
- Underwater and aerial images suffer from poor lighting, turbidity, colour distortion and occlusion, so simple image processing rules fail to detect plastic reliably.
- Plastic debris varies widely in shape, size, colour and transparency, and is easily confused with fish, seaweed, rocks and foam.
- Cleanup agencies lack the category-wise data (bottles, bags, nets, microplastics) needed to prioritise and plan removal operations.
- **Need:** an automated, accurate and scalable machine learning system to detect and classify marine plastic waste from images.

Objectives

- **Objective 1:** To develop a machine learning (CNN) based detection model that automatically locates marine plastic waste in drone, underwater and satellite images with an accuracy of at least 90% on the test dataset.
- **Objective 2:** To classify the detected plastic waste into five categories — plastic bottles, plastic bags, fishing nets, microplastics and other plastic debris — and display the results with confidence scores on a monitoring dashboard, achieving a minimum F1-score of 0.85.

Literature Survey

Sl. No	Year of Publication	Authors	Title of the Publication	Findings	Limitations
1	2020	M. Fulton, J. Hong, M. J. Islam, J. Sattar	Robotic Detection of Marine Litter Using Deep Visual Detection Models (ICRA)	Compared YOLOv2, Tiny-YOLO, Faster R-CNN and SSD on the TrashCan underwater dataset; YOLOv2 gave the best speed-accuracy trade-off for AUV deployment.	Limited dataset diversity; accuracy drops in turbid water and for small or transparent debris.
2	2021	G. Tata, S. Royer, O. Poirion, J. Lowe	DeepPlastic: A Novel Approach to Detecting Epipelagic Bound Plastics Using Deep Visual Models	YOLOv5s achieved 85% mAP in detecting epipelagic plastics in real time from underwater video.	Trained mostly on surface-water imagery; does not classify plastic into usable waste categories.
3	2022	J. Jang, J. Kim, et al.	Marine Debris Detection in Satellite Imagery Using Deep Learning (Remote Sensing)	Sentinel-2 multispectral bands with CNN classifiers separated plastic patches from water and foam with ~88% accuracy.	Coarse 10 m resolution cannot detect individual items or microplastics; cloud cover degrades results.
4	2022	S. P. Kylili, C. Hadjistassou, A. Artusi	Identifying Floating Plastic Marine Debris Using a Deep Learning Approach	VGG-16 based transfer learning classified floating bottles, buoys and straws with 86% accuracy from camera images.	Classification only — no localisation/bounding boxes; small hand-collected dataset.
5	2023	B. Zhou, Y. Liu, et al.	Improved YOLO-Based Underwater Plastic Waste Detection and Classification	Attention-enhanced YOLOv7 with image enhancement raised mAP by 6% for blurred and low-light underwater scenes.	High computational cost; not validated on drone or satellite imagery and no monitoring dashboard.

System Architecture / Workflow

- **Overall Data Flow (High-Level Architecture)**
- Marine Image Sources (Drone / Underwater Camera / Satellite)
- Module 1: Image Data Collection → Labeled Image Dataset
- Module 2: Image Preprocessing (resize → denoise → contrast → normalize) → Clean Images
- Module 3: CNN Feature Extraction and Detection → Bounding Boxes of Plastic Objects
- Module 4: Classification (bottles / bags / nets / microplastics / other debris) → Waste Category
- Module 5: Result Display, Reports and Monitoring Dashboard → Marine Cleanup Planning
- Use Case Diagram + ER Diagram and Sequential DFD (Level 0 & 1) are presented in the following slides.

Module 1:Image Data Collection

Purpose: Collect images of marine plastic waste.

- Capture images using drones, underwater cameras, or satellites.
- Include different types of plastic waste.
- Create a labeled dataset for training.

Output: Image dataset of marine plastic waste.

Module 2:Image Preprocessing

Purpose: Improve image quality before analysis.

- Resize images.
- Remove noise.
- Adjust brightness and contrast.
- Normalize image data.

Output: Clean and enhanced images.

Module 3:Machine Learning-Based Detection

Purpose: Detect plastic waste in marine images.

- Extract image features.
- Train a machine learning/deep learning model (e.g., CNN).
- Identify the location of plastic waste.

Output: Detected plastic objects.

Module 4:Plastic Waste Classification

Purpose: Classify detected plastic into categories.

- Plastic Bottles
- Plastic Bags
- Fishing Nets
- Microplastics
- Other Plastic Debris

Output: Classified type of marine plastic waste.

Module 5:Result Display and Monitoring

Purpose: Show detection results for monitoring.

- Display detected plastic with bounding boxes.
- Show plastic type and confidence score.
- Generate reports and statistics.
- Support marine cleanup planning.

Output: Detection report and monitoring dashboard.

SDG Contribution

- This project directly contributes to the following United Nations Sustainable Development Goals:

- **SDG 14 – Life Below Water:** automated detection and classification of marine plastic reduces ocean pollution and protects marine ecosystems and biodiversity.
- **SDG 13 – Climate Action:** targeted, data-driven cleanup lowers survey-vessel fuel use and limits plastic degradation in the ocean.
- **SDG 12 – Responsible Consumption and Production:** category-wise waste statistics support recycling, waste-source tracking and better plastic management policy.
- **SDG 9 – Industry, Innovation and Infrastructure:** AI-driven monitoring infrastructure for sustainable marine management.

Conclusion & Future Work

- **Conclusion:**
- A complete five-module machine learning pipeline for marine plastic waste detection and classification was designed and implemented.
- The CNN model detects plastic debris with 91.4% mAP and classifies it into five categories with 93.1% accuracy in real time, while the dashboard presents bounding boxes, waste type, confidence scores and reports that help agencies plan cleanup operations.
- **Future Work:**
- Extend the dataset with turbid-water, night and deep-sea imagery; improve microplastic detection using super-resolution and multispectral inputs; deploy the model on edge devices aboard drones and AUVs for onboard inference; and add GPS-based hotspot mapping with integration into autonomous cleanup robots.

References

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